

■ AZ-104 – LAB 04 TASK 2: Create ManufacturingVnet using ARM Template 🧠 Purpose of Task 2

Task 2 demonstrates Infrastructure as Code (IaC) by reusing an exported ARM template to deploy a new virtual network with different address spaces and subnets.

Instead of manually creating resources again, we:

Reused existing configuration

Modified values safely

Deployed consistently

This mirrors real enterprise Azure operations.

📄 What Was Implemented Virtual Network Created

Name: ManufacturingVnet

Resource Group: az104-rg4

Region: East US

Address Space: 10.30.0.0/16

Subnets Created Subnet Name Address Range Purpose SensorSubnet1 10.30.20.0/24 Manufacturing sensors
SensorSubnet2 10.30.21.0/24 Additional sensor workloads 🧩 Key Technical Steps (What You Actually Did) **1**
Reused ARM Template

Exported from CoreServicesVnet

Modified instead of rebuilding manually

2 Address Space Isolation

CoreServicesVnet → 10.20.0.0/16

ManufacturingVnet → 10.30.0.0/16

✓ Prevents overlap ✓ Enables future VNet peering / hybrid connectivity

3 Subnet Renaming & Readdressing

Logical subnet names aligned with manufacturing workloads

Ensures clarity and scalability

4 Deployment via Azure Portal

Used Deploy a custom template

Loaded:

template.json

parameters.json

Verified deployment success

🔍 Validation Performed

ManufacturingVnet visible in portal

Correct address space applied

Subnets created with expected CIDR ranges

No overlap with CoreServicesVnet

🧠 Concepts Covered (Important for Interviews)

Infrastructure as Code (ARM Templates)

Reusable Azure deployments

CIDR planning across environments

Template-based automation

Safe modification of production templates

🔗 Interview-Ready Questions & Answers

Q: Why use ARM templates instead of portal deployment? A: ARM templates provide consistency, automation, and repeatability.

Q: Why isolate VNets using different address spaces? A: To prevent routing conflicts and support future connectivity.

Q: What is the advantage of reusing templates? A: Faster deployments and reduced human error.

Q: Can ARM templates be deployed multiple times? A: Yes, they are idempotent and reusable.

☑ Task 2 Key Takeaways

ARM templates are central to Azure automation

Address planning must consider future connectivity

Templates allow rapid scaling with minimal effort

Portal-created resources can always be converted to code